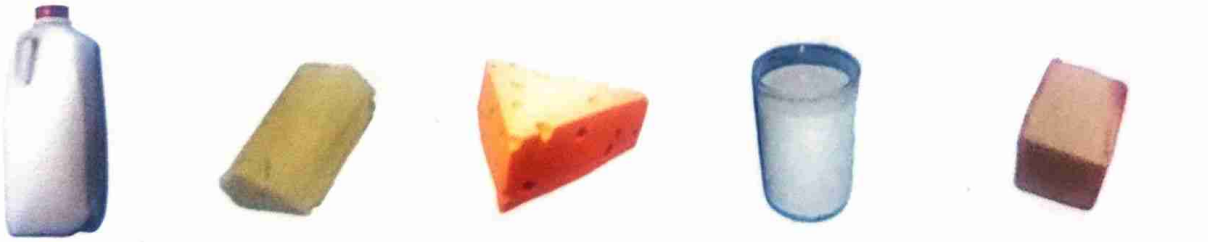
HUSBANDRY AND ECONOMICS OF DAIRY CATTLE, GOATS AND MILK PROCESSING MANUAL

HEIFER INTERNATIONAL







MODULE IV: MILK PROCESSING AND HANDLING

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TABLE OF CONTENTS

LIST OF TABLES 179

LIST OF FIGURES 179

ACRONYMS 180

LEARNING OBJECTIVES OF THE MILK PROCESSING MODULE 181

1. INTRODUCTION 182

2. MILK COMPOSITION AND STRUCTURE 183
   - 2.1 What is Milk? 183
   - 2.2 What is the structure of milk? 183
   - 2.3 What is the general composition of milk? 183
   - 2.4 What are the specific constituents of milk? 184
     - 2.4.1 Milk fat 184
     - 2.4.2 Milk Proteins 185
     - 2.4.3 Carbohydrates 185
     - 2.4.4 Minerals and Vitamins 185
     - 2.4.5 Water 185
   - 2.5 What factors affect the composition of milk? 185
     - 2.5.1 Genetic factors 186
     - 2.5.2 Environmental factors 186
   - 2.6 General properties of milk 186

3. MILK MICROBIOLOGY 187
   - 3.1 Introduction 187
     - 3.1.1 What is Microbiology? 187
     - 3.1.2 Important microorganisms in milk 188
   - 3.2 Sources of contamination 188
   - 3.3 Growth of microorganisms 188
   - 3.4 Effects of microbial growth in milk 189
   - 3.5 Mastitis and Somatic Cells 189
     - 3.5.1 What is Mastitis? 189
     - 3.5.2 What are Somatic Cells? 189
   - 3.6 Methods for the destruction (killing) of microorganisms in milk 189
     - 3.6.1 Milk Pasteurization 190
     - 3.6.2 Milk Sterilization 190

4. FOOD QUALITY AND FOOD SAFETY 191
   - 4.1 Definitions 191
     - What is food safety? 191
     - What is quality or food suitability? 191
     - What is Food Hygiene? 191
     - What are hazards? 191
   - 4.2 Food safety control 191
   - 4.3 Activities in the milk processing chain 192
   - 4.4 What happens at the milk collection centre? 193
   - 4.5 Why does the dairy industry pay me less for the milk supplied? 193
   - 4.6 Laws, Regulations and Regulatory Bodies governing milk production and processing in Zambia 194
     - Food Act of Zambia 194
     - Zambia Bureau of standards 194
     - Codex alimentarius 195
     - Zambia Dairy Processors Association (ZDPA) 195
   - 4.7 Involvement of the Government, Industry and the Consumer in food safety aspects ... 195
     - 4.7.1 Government 195
     - 4.7.2 Industry / Trade 196
     - 4.7.3 Consumers 196

LEARNING OBJECTIVES OF THE MILK PROCESSING MODULE

Trainees should be able to:

Section 2: Milk composition and structure

- Outline the basic composition and structure of milk

Section 3: Milk microbiology

- State the groups of microorganisms commonly found in milk and milk products, and sources of such groups of microorganisms
- Give basic explanations on the effect of the presence of microorganisms on milk and milk products
- State the factors that encourage and inhibit the growth of such microorganisms in milk and milk products, and how the participants can use such knowledge in milk and milk products processing and storage

Section 4: Food quality and Food Safety

- Basically define food quality and food safety
- Identify the categories of hazards associated with milk and milk products
- Outline the basic food quality and food safety control measures in a milk chain
- Explain the relationship between the level of milk quality and/or safety, and the remuneration for the supply of fresh raw milk to buyers
- Mention the basic milk quality and safety infrastructure existing in Zambia

Section 5: Hygienic milk handling

- Explain how to practically handle fresh raw milk hygienically right from the stable/parlor up to the transportation of such milk to the market

Section 6: Milk processing

- Explain how to practically process fresh raw milk into pasteurized milk products (whole and skimmed), cream-products, butter-products, cheese-products and yogurt-products on a small scale level
- Explain how to practically process sour raw milk into sour butter and cottage cheese on a small scale level
- Explain how to deal with some expected quality and safety problems during the processing of fresh raw milk and sour raw milk into their respective products on a small scale level
- Give basic explanations of the effects of processing and storage conditions on milk and milk products in relation to the basic composition and structure

1. INTRODUCTION

In Zambia, milk is produced by both subsistence pastoralists and commercial farmers. It is either sold fresh or consumed as fermented milk (yoghurt and sour milk) or processed into products such as butter, and cheese. Sour milk (Mabisi) is the most common product in cattle rearing communities. Milk is usually soured before any further processing is done and in the case of the rural setup the milk sours naturally due to lack of cold storage facilities. There are few milk processing plants in Zambia and usually these plants purchase the milk from commercial farmers and a few small scale farmers who are organized in cooperative groups. The majority of the upcoming small scale farmers still face marketing problems because their milk quantities are low to sell to larger processors as an individual dairy farmer. However, in most districts the milk collection centers have not been established in order for the milk companies to purchase the milk in bulk. There is therefore need for milk produced by rural smallholders to be processed on the farm using 'traditional technology'. It is imperative, therefore, to consider these processes and look for possible technological interventions at this scale when considering dairy development in the rural sector in Zambia.

Very few farmers (both commercial and small scale) in Zambia produce sour milk, butter and cottage cheese for sale. It is worth noting that while the processes that they use have not been subjected to extensive scientific investigation; they appear to be effective methods of converting milk into stable marketable products. Milk is processed primarily to convert it into more stable products; for example, fermented milk can be stored for up to 20 days (O'Connor, 1995). Milk products are more stable compared to fresh milk because they are more acidic and/or contain less moisture. Preservatives may also be added. Thus, by increasing acidity and reducing moisture content, the storage stability of milk can be increased (O'Connor, 1995).

This module deals with rural milk processing. It concentrates on 'traditional' products or on products that can be easily made, need little specialized equipment and that are easily adaptable to the rural processing plants. Topics on milk composition and structure, milk microbiology, hygienic milk handling and food safety have also been covered.

2. MILK COMPOSITION AND STRUCTURE

2.1 What is Milk?

Milk is an opaque white liquid produced by the mammary glands of female mammals. The role of milk in nature is to nourish and provide immunological protection for the mammalian young (Wong, 1988). It is not surprising, therefore, that the nutritional value of milk is high.

For technological purposes, milk may be defined as an emulsion of butterfat globules within a water-based fluid.

Fat globules are surrounded by a membrane (emulsifer). The fat in the milk is in the form of an emulsion; that is, a suspension of small liquid globules that do not mix with the water in milk. The stability of the fat emulsion is overcome by allowing milk to stand for a period of time, which results in the cream rising to the top. This is especially true for high fat milk.

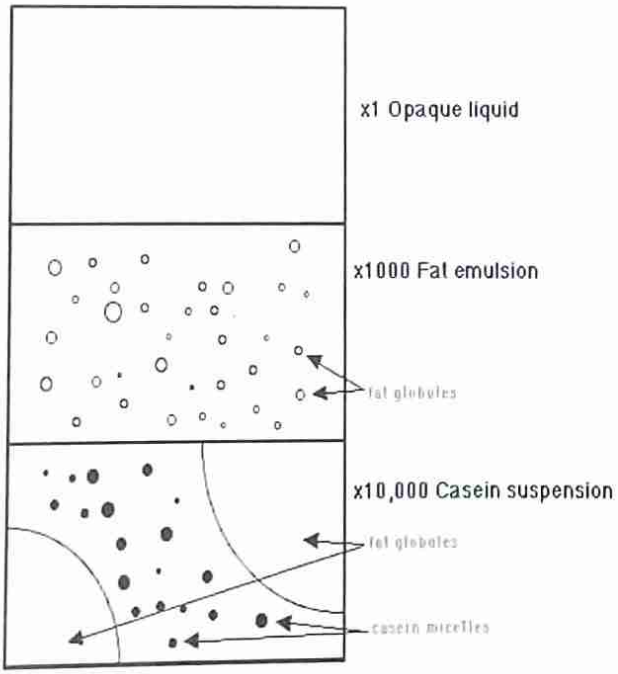
This section outlines the structure, composition of milk, factors affecting composition of milk and the general properties of milk.

2.2 What is the structure of milk?

- Whole milk
- Milk fat (Fat globules) phase
- Water phase (plasma)
- Milk proteins (caseins/whey)

❖ Milk minus fat = plamsa (skimmed milk)
❖ Milk plasma minus caseins = serum

Milk Structure



2.3 What is the general composition of milk?

When all other factors are considered, only an approximate composition of milk can be given as shown in Table 2.1

Table 2.1 General composition of cow's milk

Constituent	Range (%)	Average
Water	85.5 – 88.7	87.3
Total solids	10.5 – 14.5	13.0
Fat	2.5 – 6.0	4.0
lactose	3.6 – 5.5	4.8
proteins	2.9 – 5.0	3.4
minerals	0.6 – 0.9	0.8
Solids not fat	7.9 – 10.0	8.8

Source: O'Connor, 1995

Given the above general composition of cow milk, some relationships between constituents are very stable and can be used to indicate whether any tampering with the milk composition has occurred.

2.4 What are the specific constituents of milk?

Milk is composed of milk fat, proteins, carbohydrates, vitamins, minerals and water.

2.4.1 Milk fat

- ⇒ Milk fat consists of a large number of spherical microscopic globules of varying sizes floating in the milk.
- ⇒ Each globule is surrounded by a thin skin - the fat globule membrane - which acts as the emulsifying agent for the fat suspended in milk.
- ⇒ The membrane protects the fat from enzyme attack and prevents the globules coalescing into butter grains.
- ⇒ The fat is present as an oil-in-water emulsion that can be broken by mechanical action such as homogenization (shaking).

At room temperature, the lipids are solid, therefore, are referred to as "fat" as opposed to "oil" which is liquid at room temperature. The final melting point of milk fat is 37˚C. This temperature is important because 37˚C is the body temperature of the cow and the milk would need to be liquid at this temperature.

How can milk fat get spoiled?

(a) Hydrolytic rancidity

Lipolysis is the breaking down of milk fat into component fatty acids. Lipolysis is induced by the action of naturally occurring lipase enzyme in milk. It increases the concentration of free fatty acids.The susceptibility of milk to lipolysis is increased by homogenisation, pumping, foaming and temperature manipulation.

Don't
Avoid alternate heating and cooling of milk.

(b) Oxidative rancidity

The oxidative deterioration of lipids is caused by oxidation (involving oxygen) of unsaturated fatty acids, resulting in the production of off-flavors.

Factors influencing the rate of fat oxidation: high temperature, light and trace elements (copper, iron etc).

Do
Exclusion of oxygen, refrigeration and packaging in opaque or colored containers

2.4.2 Milk Proteins

Cow milk contains about 34 g/litre protein. Of this, about 27 g/litre consists of caseins which are precipitated upon acidification to pH 4.6 at temperatures above 20°C. The proteins remaining in solution at pH 4.6 are called whey proteins

Minor protein constituents
A number of enzymes have been detected in bovine milk. Some milk enzymes act on substrates present as normal constituents of milk and may play either beneficial or deleterious roles during milk processing (See Table 2.2 in Annex 1).

2.4.3 Carbohydrates

Lactose is the major carbohydrate fraction in milk. It is a disaccharide composed of two sugars, glucose and galactose.

⇒ The average lactose content of milk varies between 3.6 and 5.5%.

⇒ Mastitis reduces lactose content in milk.

⇒ Lactose is a source of energy for the young calf and provides 4 calories/g of lactose metabolized.

⇒ Lactose is less soluble in water than sucrose and is also less sweet.

⇒ Milk contains other carbohydrates in small amounts, including glucose, galactose and oligosaccharides.

Don't
Avoid prolonged heating of milk above 100°C because it causes lactose to combine irreversibly with the milk proteins. This reduces the nutritional value of the milk and also turns it brown.

2.4.4 Minerals and Vitamins

⇒ Milk contains the fat-soluble vitamins A, D, E and K in the fat fraction and water-soluble vitamins B complex and C in the water phase.

⇒ Vitamins are unstable and processing can therefore reduce the effective vitamin content of milk.

⇒ All minerals considered essential to the human diet are present in milk. However, the major minerals in milk are calcium and phosphorous.

⇒ Other elements in smaller quantities include copper, iron, lead, boron, manganese and iodine.

2.4.5 Water

⇒ Cow milk is about 87% water, so the transportation of milk from dairy farms involves hauling of considerable amounts of water.

⇒ In all animals, water is the nutrient required in the greatest amount and milk does supply a great amount of water.

⇒ Milk production is very rapidly affected by a shortage of water and drops the same day drinking water is limited or unavailable.

2.5 What factors affect the composition of milk?

The composition of milk varies considerably due to genetic and environmental factors (O'Connor, 1995). The quantities of the main milk constituents (fat, protein, lactose) can also vary considerably depending on the individual animal, its breed, stage of lactation, age and health status.

2.5.1 Genetic factors

The **breed and the species** of the dairy animal will influence the composition of the milk.

- Jersey and Guernsey breeds give milk with about 5% fat
- Shorthorns and Friesians contain about 3.5% fat.
- Goat milk has relatively higher quantities of protein and fat, 3.5% and 4.3%, respectively, compared to cow milk. Table 2.2 gives average compositions of milk from different animals.

Table 2.2 **Average Composition of milk from different animals**

Nutrient (g/100g)	Cow	Buffalo	Human	Goat
Water	87.3	84.0	87.5	86.8
Protein	3.4	3.7	1.0	3.5
Fat	4.0	6.9	4.4	4.5
Lactose	4.8	5.2	6.9	4.7
Minerals	0.8	0.8	0.2	0.8

Source: Wong, 1988; O'Connor, 1995

2.5.2 Environmental factors

Different factors around the dairy animal will influence the composition of the milk produced:

⇒ **Stage and number of lactation** will have an influence on the structure of the milk. If the cow is milked immediately after parturition, the milk called colostrum differs from the normal milk. Colostrum contains more mineral salts and protein and less ash than later milk (Wong, 1988). The difference in the composition of the milk is because the needs of the calf differ at different stages.

⇒ The **interval between milking** will affect the composition mostly due to the time between the milking. Morning milk will have more fat due to the longer time from evening milking to morning milking.

⇒ The **age** and **health of the animal** will also affect the composition of milk. As the animal ages, the milk fat content decreases. When an animal has mastitis, the milk composition will also change.

⇒ The **nutrition** of the animal impacts on the quantity and quality of the milk. A well fed animal will produce more milk of a higher quality than an underfed one.

2.6 General properties of milk

Some of the important properties of milk include

- Density
- Freezing point
- Acidity

3. MILK MICROBIOLOGY

3.1 Introduction

3.1.1 What is Microbiology?

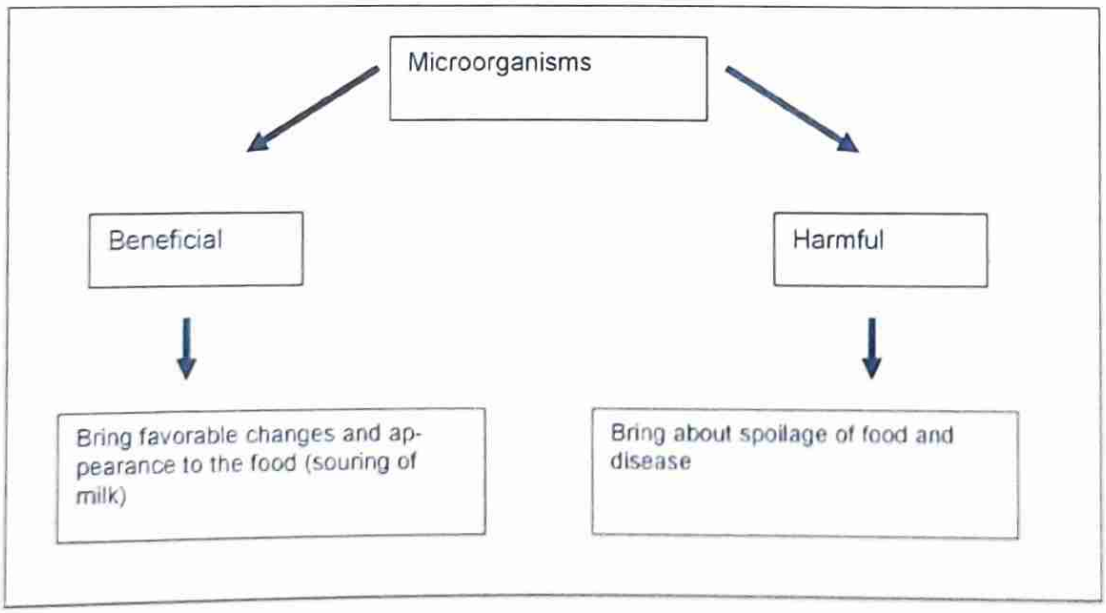
⇒ Study of organisms too small to be seen with the naked eye.
⇒ The Small organisms are known as microorganisms.
  - They are found everywhere (ubiquitous)

⇒ Normal Microflora is the mixture of organisms regularly found at any anatomical site:
  - Surface of leaves, flowers, fruits and especially the roots,
  - Surfaces of animals and the stomach (gut) of animals

⇒ Normal flora may affect:
  - Quality of raw material used in manufacture of food
  - The kind of contamination (which may occur during processing)
  - Spoilage and food associated illnesses

⇒ Milk is a nutritious food for human consumption
  - Has available nutrients
  - High water content
  - Moderate pH (6.4 to 6.6)

⇒ Therefore its also good for growth of microorganisms
⇒ Milk from a healthy cow is sterile at secretion into the udder
⇒ Before it leaves the udder it contains low numbers of microorganisms
⇒ Microorganisms can either be beneficial or harmful (See diagram below).



3.1.2 Important microorganisms in milk

For details of this section see annex 1

3.2 Sources of contamination

Milk may become contaminated with microorganisms particularly bacteria during or after milking. Table 3.1 gives the different sources of contamination of milk with microorganisms

Table 3.1 Sources of Contamination

Sources	Comments
The udder of the dairy animal	The interior of the teats of the udder is warm and contains last remains of milk, such that when the microorganisms enter through the opening of the teat they can multiply.
The skin of the dairy animal	Soil, faeces and dirt adhere to the skin and hair of the dairy animal. Hair, dirt and dust fall into milking utensils. Faeces contain large amounts of microorganisms and most of them are pathogenic.
Utensils and Equipment	Milking utensils and equipment are the major sources of contamination of milk. The soil of previous activities can be a source of microorganisms and also nutrients for microorganisms if not properly washed.
Feed	Microorganisms are found everywhere and are more abundant in vegetation and soil. Dry feed have more bacteria and less fungi. These microorganisms contaminate milk.
Air of the dairy animal shed	The air of the animal shed is greatly contaminated by dry dirt and dust. During the mixing of feeds and during cleaning process of the floor, the air of the animal shed is highly contaminated and it can be passed on to the milk.
Milking persons	Pathogenic microorganisms may enter into the milk through milking persons. The milking persons' dirty clothes and improperly washed hands and unclean nails can also be a source of microorganisms. Discharge from sneezing, coughing and nose blowing which contain a lot of microorganisms should not reach the surrounding atmosphere, milking utensils and equipment or the milk.
Water.	Pure water should be used for cleaning purposes. Water exposed to contamination spreads the microorganisms. Chlorination of water prevents such contamination.

NOTE:
Once microorganisms get into the milk, their numbers increase rapidly. It is more effective to exclude microorganisms than to try to control microbial growth once they have entered the milk, because for most microorganisms their generation time (time it takes for one bacterial population to double) is short.

3.3 Growth of microorganisms

Microorganisms multiply during production and holding of milk, depending on storage time and other conditions. For the stages and factors influencing growth of microorganisms see 3.4 in annex 1.

3.4 Effects of microbial growth in milk

Souring	Most common, due to transformation of lactose into lactic acid and other compounds, principally by lactic acid bacteria
Aroma Production	Due to production of desirable flavor compounds
Proteolysis	Protein decomposition leading to unpleasant odor
Ropiness	Long threads of milk are formed while pouring

3.5 Mastitis and Somatic Cells

3.5.1 What is Mastitis?

⇒ Inflammation of the udder, typically caused by a microbiological infection.

⇒ Many types of microorganisms can cause infection

⇒ The microorganisms can enter the udder and multiply and subsequently contaminate milk

⇒ Mastitis in dairy animals can be a source of disease causing bacteria (pathogens) and spoilage organisms in milk

⇒ Milk from mastitic dairy animals

- may have off-flavors
- may undergo deterioration of the milk fat and protein more quickly than milk from healthy animals.

Dos

⇒ Animals found with mastitis should be separated from the healthy animals to:

- help control spread of infection
- ensure the quality of milk produced on the farm
- Treat animals with antibiotics

⇒ Milk from treated animals should either be:

- discarded or put separate to avoid contamination of milk from healthy animals

⇒ Milk from animals treated with antibiotics should not be used for human consumption

3.5.2 What are Somatic Cells?

⇒ Somatic cells are cells from the animal predominantly white blood cell (leukocytes) present in milk.

⇒ During most mastitis infections, the number of somatic cells present in the udder increases to help the cow fight the infection.

⇒ An increase in somatic cells count during a mastitis infection increases the amount of destructive enzymes present in milk, which increases the rate of deterioration of milk fat and protein.

3.6 Methods for the destruction (killing) of microorganisms in milk

Heat is most widely used.

3.6.1 Milk Pasteurization

⇒ Most common process used to destroy bacteria in milk.

⇒ In pasteurisation, the milk is heated to a temperature sufficient to kill pathogenic bacteria, but well below its boiling point.

⇒ This also kills many non-pathogenic microorganisms and thereby extending the storage stability of the milk

3.6.2 Milk Sterilization

⇒ Milk is subjected to severe heat treatment that ensures almost complete destruction of the microbial population.

⇒ The product is then said to be commercially sterile.

⇒ Time/ Temperature treatments of above 100°C for 15 to 40 minutes are used.

⇒ Product has a longer shelf life than pasteurized milk.

⇒ Another method of sterilisation is ultra-heat treatment, or UHT. In this system, milk is heated under pressure to about 140°C for 4 seconds.

⇒ The product is virtually sterile. However, it retains more of the properties of fresh milk than conventionally sterilised milk.